

第 1 课 传输介质

一、单项选择题

1. Which is one of the key features of Global Positioning System (GPS)?
A. Accuracy about 2 kilometers B. 1200 total satellites orbit the earth.
C. Provide accurate time information. D. Open to American users only.
2. Which of the following energies can traverse long distances and penetrate objects such as the walls of a building?
A. electrical B. heat C. light D. radio
3. In a factory that uses electric arc welding equipment, () is the BEST wiring for short-distance communication.
A. unshielded twisted pair B. shielded twisted pair
C. coaxial cable D. optical fibers
4. To reduce the interference caused by random electromagnetic radiation, communication systems DO NOT use () .
A. twisted pair wiring B. parallel pair wiring
C. coaxial cable D. none of above
5. Which of the following media is used in RJ-45 wire cable?
A. Twisted pair wire. B. Parallel pair wire. C. Coaxial cable. D. Fiber.
6. Communication to a satellite uses () energy.
A. light B. heat C. radio D. electrical
7. Optical fiber is more desirable than copper wiring, EXCEPT:
A. Resistance to electrical noise. B. Higher bandwidth.
C. Lower overall cost. D. Less signal energy loss.
8. () may NOT be sufficient to run above the ceiling in an office building on top of a florescent light fixture.
A. Twisted pair wiring B. Coaxial cable C. Shielded twisted pair D. None of above

9. RJ-45 network cable with which we connect two computers uses () cable.
- A. parallel-pair B. twisted-pair C. coaxial D. fiber-optic
10. The () is not affected by electromagnetic interference or noise.
- A. satellite B. twisted-pair C. coaxial D. optical fiber
11. Which of the following can be used with especially strong electrical noise?
- A. parallel-pair B. twisted-pair C. coaxial D. none of above
12. Which of the following technologies is used in 4G network?
- A. radio B. twisted-pair C. coaxial D. optical
13. 关于单模光纤，下面的描述中错误的是 ()。
- A. 芯线由玻璃或塑料制成 B. 比多模光纤芯径小
- C. 光波在芯线中以多种反射路径传播 D. 比多模光纤的传输距离远

二、简答题

14. 根据项目一（或实验一）利用可见光近距离传输，回答问题：（1）项目使用的是模拟信号还是数字信号？（2）结合通信模型中信源、编码、信道、解码、信宿和噪声的概念，分析该系统，指出其中的各项的具体内容是什么？（3）该项目是否引导型传输媒介？
15. 设计网络最基本的内容是选用合理的传输介质，按能量可以分为电气、光和无线电波等三大类，请列举常用的 4 种传输介质，分别指出属于哪一类。
16. 选择传输介质需要权衡哪些方面？试其解释为何企业内部网络布线多用导线，而校区和校区之间多使用光纤。

三、编程题

17. 用两台笔记本电脑相对以模拟光通信。其中，显示器模拟发光二极管，摄像头模拟光敏感应器。请编写 C++ 程序利用 OpenCV 库实现：
- （1）假设我们用黑色和白色分别代表比特 1 和 0，请你写一个程序控制显示器的颜色，另一个程序识别该颜色所表示的比特。
- （2）假设我们用黑、白、红、蓝、绿、紫、黄、青表示八进制位 0、1、……、7（请设计一种便于实现的方案）。

分别为两题各写一个接口：Scalar encode(int msg);和 int decode(Scalar color);;
为两题写一组共同的接口：void send(int msg);和 int receive();。

如有条件，可以完善可执行程序流程，使用两台笔记本模拟，或者用生成和解析图像文件模拟，验证其正确性。